

SML Assignment 4 - Final Explained Report

Q1 AdaBoost (Detailed Explanation)

Approach: MNIST dataset was filtered for classes 4 and 9 and relabeled as -1 and 1. Data was flattened, normalized, and reduced to 5 dimensions using PCA computed only on training data. Decision stumps were used as weak learners, selecting optimal feature and threshold minimizing weighted error.

Formulas Implemented:

Weighted error: $\epsilon = \sum w_i I(y_i \neq h(x_i))$

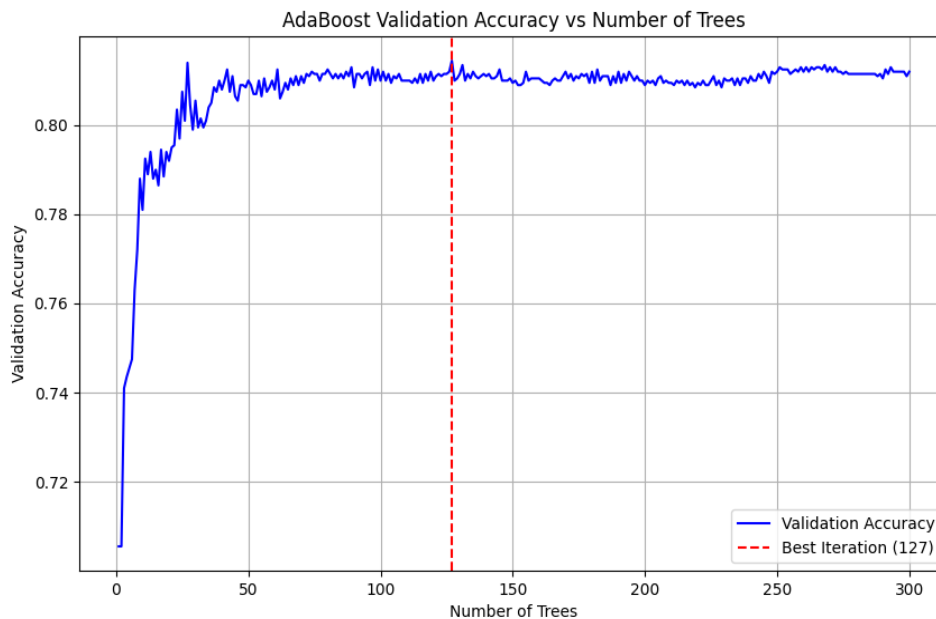
Alpha: $\alpha = 0.5 \ln((1-\epsilon)/\epsilon)$

Weight update: $w_i \leftarrow w_i \exp(-\alpha y_i h(x_i))$

Additional Detail: Candidate thresholds were midpoints of sorted unique values (random 1000 sampled per feature). 300 stumps were trained sequentially.

Results: Best iteration = 127, Validation Accuracy = 81.45%, Test Accuracy = 82.77%

Plot Insight: Accuracy sharply increases (~70%→80%) in first ~40 trees, then saturates near ~82%. Peak at 127 confirms optimal ensemble size.



Conclusion: AdaBoost improves performance by focusing on hard samples, but optimal stopping is required.

Q2 Gradient Boosting (Detailed Explanation)

Approach: Regression stumps were trained sequentially to minimize SSR using pseudo-residuals derived from absolute loss. Labels updated iteratively.

Formulas:

Residual: $r_i = y_i - F(x_i)$

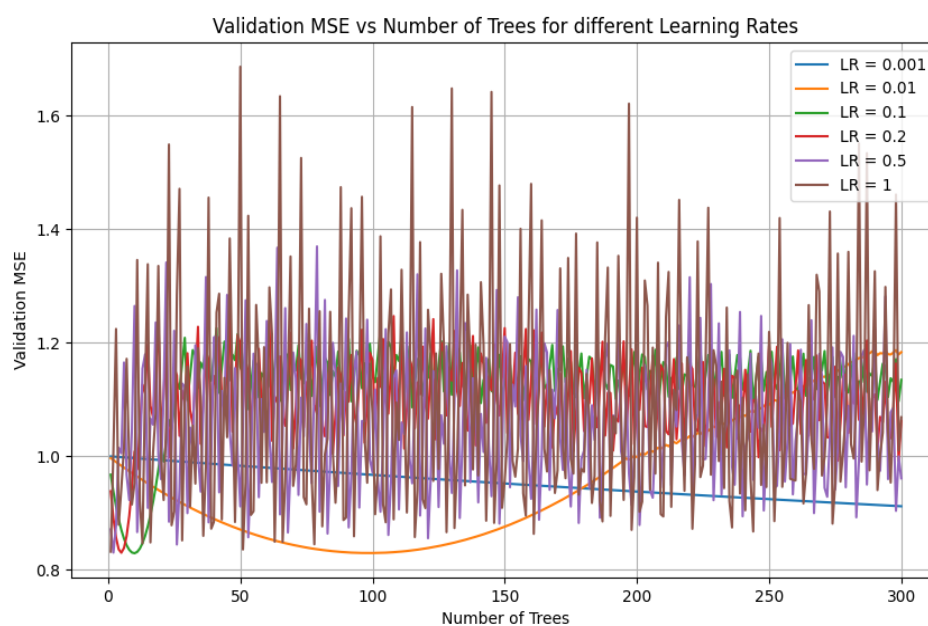
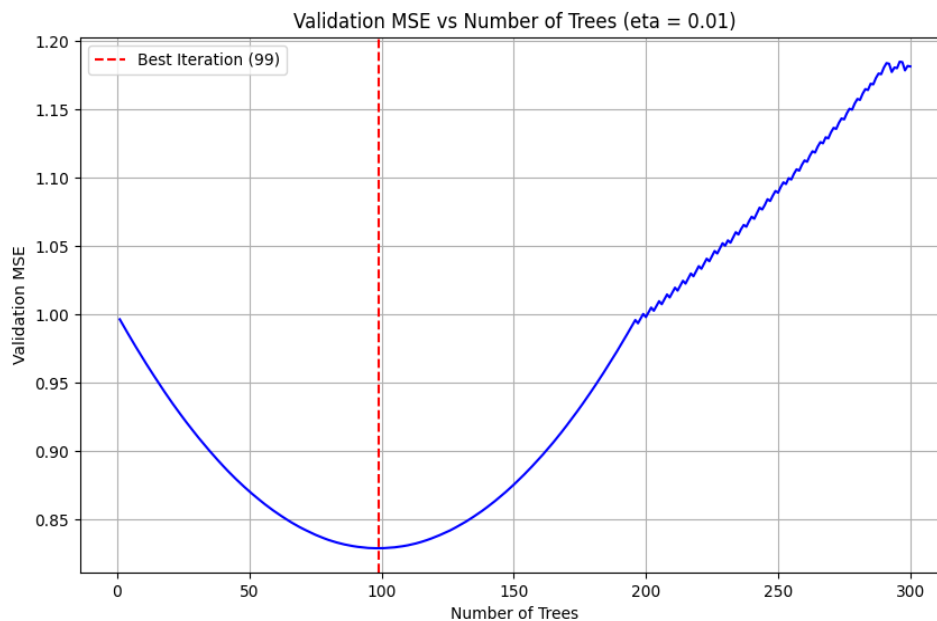
Update: $F_m(x) = F_{m-1}(x) + \eta h_m(x)$

Absolute loss gradient: $\text{sign}(y - F(x))$

Additional Detail: Learning rates tested: {0.001,0.01,0.1,0.2,0.5,1}. Validation MSE tracked per iteration.

Results: Best iteration ≈ 99 ($\eta=0.01$), Validation MSE ≈ 0.83 , Test MSE ≈ 0.84

Plot Insight: MSE decreases till ~ 100 trees then increases (U-shape). Higher η shows oscillations ($\sim 0.9\text{--}1.6$), lower η smooth decline.



Conclusion: Gradient boosting follows bias-variance tradeoff; optimal learning rate ensures stable convergence.

Q3 Perceptron (Detailed Explanation)

Approach: Synthetic Gaussian datasets generated. Perceptron initialized with zero weights and trained using online updates with early stopping.

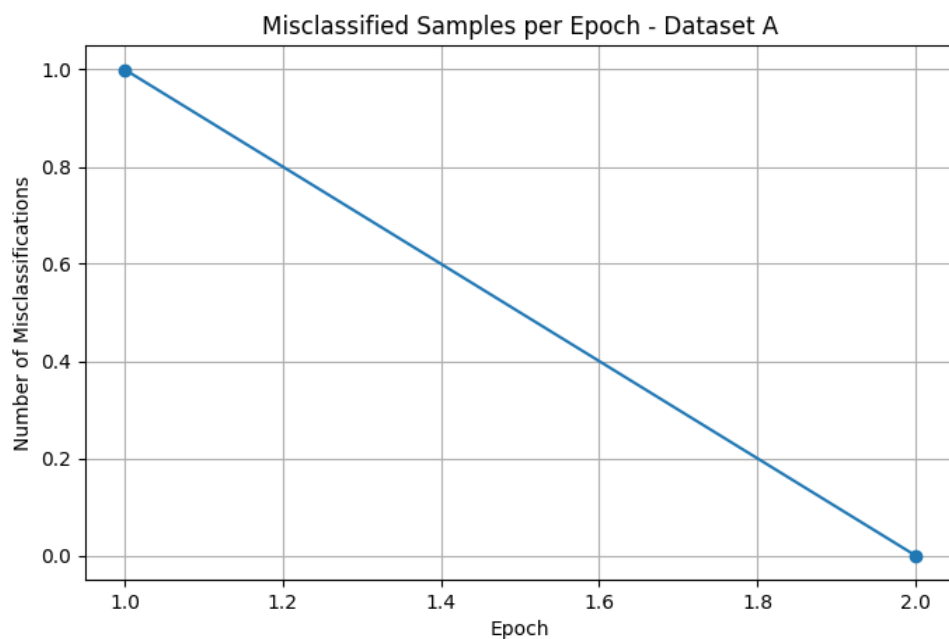
Formula:

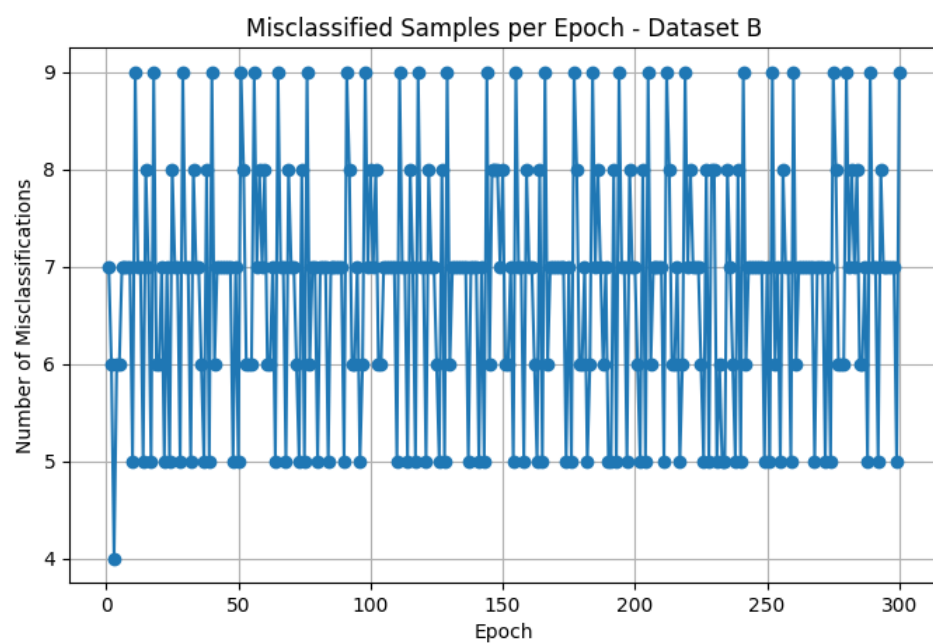
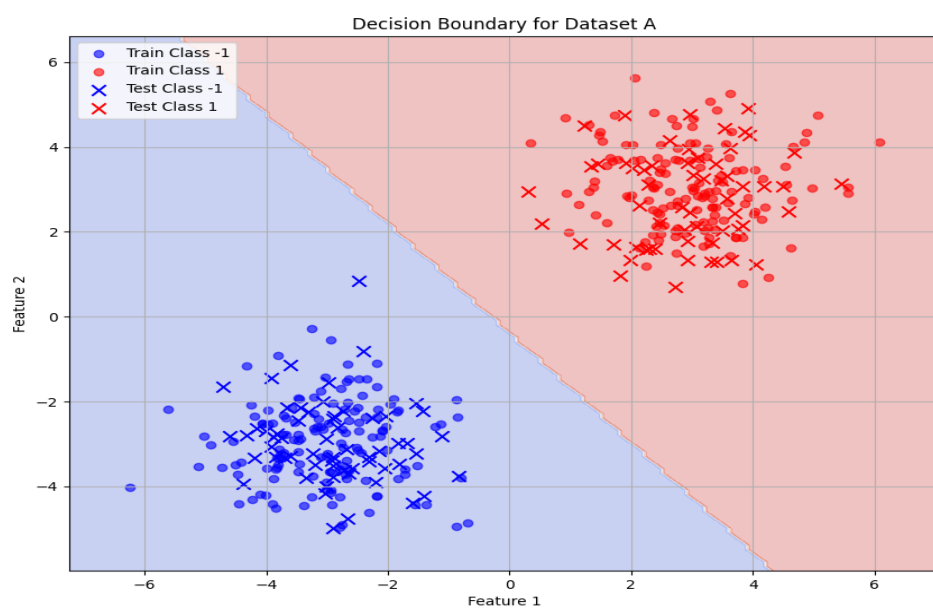
$$w \leftarrow w + \eta y x, b \leftarrow b + \eta y$$

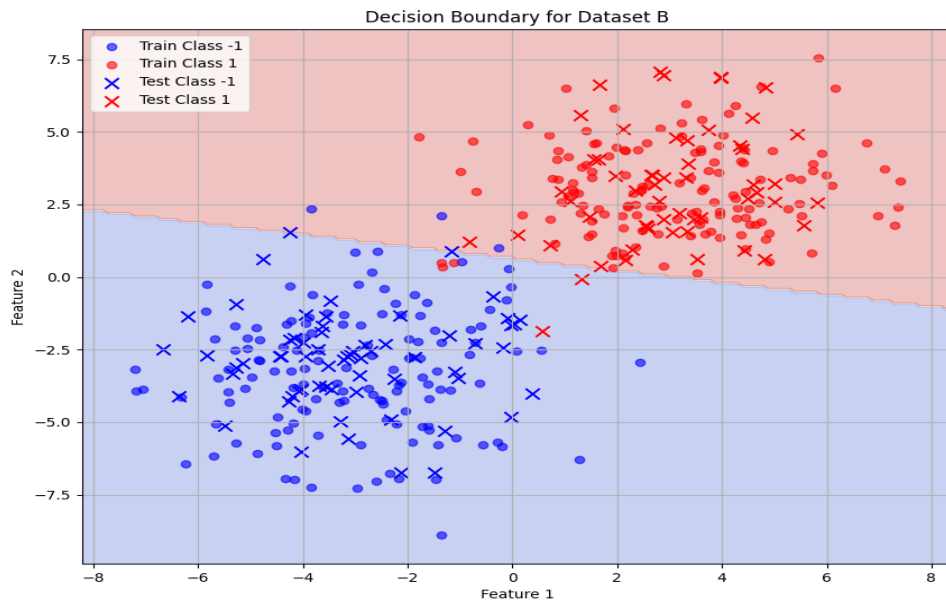
Additional Detail: Dataset A uses covariance I (separable), Dataset B uses 3I (overlap).

Results: Dataset A: converged in 2 epochs, accuracy 100%. Dataset B: no convergence (300 epochs), accuracy 96.67%

Plot Insight: Dataset A misclassification drops to 0 instantly; Dataset B fluctuates (~5–9 errors), showing non-separability.







Conclusion: Perceptron converges only for linearly separable data as per convergence theorem.